

$$\int_1^{+\infty} \frac{1}{n^\alpha} \begin{cases} \text{div se } \alpha \leq 1 \\ \frac{1}{\alpha-1} \text{ se } \alpha > 1 \end{cases}$$

$$\int_0^{+\infty} e^{\beta n} \begin{cases} \text{div se } \beta \geq 0 \\ -\frac{1}{\beta} \text{ se } \beta < 0 \end{cases}$$

Ficha 3

③ Natureza de

$$\begin{aligned} \text{a) } \int_0^{+\infty} \frac{5}{4+n^2} &= \lim_{t \rightarrow +\infty} 5 \int_0^t \frac{1}{4+n^2} = \lim_{t \rightarrow +\infty} 5 \int_0^t \frac{1}{4+n^2} = \lim_{t \rightarrow +\infty} \frac{5}{4} \int_0^t \frac{1}{1+\left(\frac{n}{2}\right)^2} = \lim_{t \rightarrow +\infty} \frac{5}{4} \int_0^t \frac{2}{1+u^2} du \\ &= \lim_{t \rightarrow +\infty} \frac{10}{4} \int_0^t \frac{1}{1+u^2} du = \frac{10}{4} \lim_{t \rightarrow +\infty} \left[\arctg(u) \right]_0^t = \end{aligned}$$

$$u = \frac{n}{2} \quad (=) \quad 2u = n \quad (=) \quad 2 du = dn$$

$$\begin{aligned} &= \frac{5}{2} \lim_{t \rightarrow +\infty} \left[\arctg\left(\frac{n}{2}\right) \right]_0^t = \frac{5}{2} \lim_{t \rightarrow +\infty} \left(\arctg\left(\frac{t}{2}\right) - \arctg(0) \right) \\ &= \frac{5}{2} \left(\frac{\pi}{2} - 0 \right) = \frac{5\pi}{4} \end{aligned}$$

$$\begin{aligned} \text{b) } \int_{\pi}^{+\infty} \cos(3n) dn &= \lim_{t \rightarrow +\infty} \int_{\pi}^t \cos(3n) dn = \lim_{t \rightarrow +\infty} \frac{1}{3} \left[\sin(3n) \right]_{\pi}^t = \\ &= \lim_{t \rightarrow +\infty} \frac{1}{3} \left(\sin(3) - \sin(3t) \right) \text{ Div.} \end{aligned}$$

$$c) \int_{-\infty}^2 \frac{1}{(4-n)^2} dn = \lim_{t \rightarrow -\infty} \int_t^2 \frac{1}{(4-n)^2} dn = \quad u = 4-n \Leftrightarrow 4-u = n \Leftrightarrow -du = dn$$

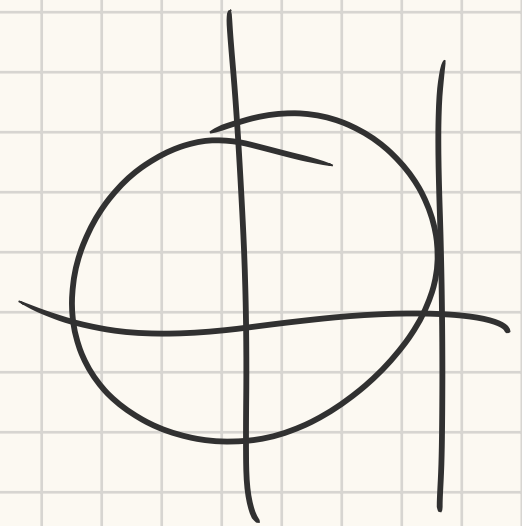
$$= \lim_{t \rightarrow -\infty} - \int_t^2 \frac{1}{u^2} du = \lim_{t \rightarrow -\infty} - \int_t^2 u^{-2} du = \lim_{t \rightarrow -\infty} - \left[\frac{u^{-1}}{-1} \right]_t^2 = \lim_{t \rightarrow -\infty} \left[\frac{1}{u} \right]_t^2$$

$$= \lim_{t \rightarrow -\infty} \left[\frac{1}{4-n} \right]_t^2 = \lim_{t \rightarrow -\infty} \left(\frac{1}{2} - \frac{1}{4-t} \right) \underset{0}{=} \frac{1}{2}$$

$$d) \int_{-\infty}^{+\infty} n \, dn = \int_{-\infty}^0 n \, dn + \int_0^{+\infty} n \, dn = \lim_{t \rightarrow -\infty} \left[\frac{n^2}{2} \right]_t^0 + \lim_{t \rightarrow +\infty} \left[\frac{n^2}{2} \right]_0^t =$$

$$= 0 - \infty \quad \text{Div}$$

$$e) \int_{-\infty}^{+\infty} \frac{1}{1+n^2} dn = \int_{-\infty}^0 \frac{1}{1+n^2} dn + \int_0^{+\infty} \frac{1}{1+n^2} dn$$



$$\text{C.A.} \quad - \int_{-\infty}^0 \frac{1}{1+n^2} dn = \lim_{t \rightarrow -\infty} \left[\arctg n \right]_t^0 = \lim_{t \rightarrow -\infty} \arctg 0 - \arctg t = 0 - \frac{\pi}{2} = -\frac{\pi}{2}$$

$$\int_0^{+\infty} \frac{1}{1+x^2} dx = \arctg(+\infty) - \arctg(0) = \frac{\pi}{2}$$

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx = \pi$$

6) Naturaleza de :

$$a) \int_1^{+\infty} \frac{\sin^2 x}{x^{5/2}} dx$$

$$\frac{\sin^2 x}{x^{5/2}} \leq \frac{1}{x^{5/2}} \text{ e como } \frac{1}{x^{5/2}} \text{ (do tipo } \frac{1}{x^\alpha} \text{ com } \alpha > 1) \text{ converge}$$

então $\int_1^{+\infty} \frac{\sin^2 x}{x^{5/2}} dx$ também converge

$$b) \int_1^{+\infty} \frac{2x}{e^{2x} - 1} dx = \int_1^{+\infty} \frac{2x}{e^{2x} - 1} dx$$

c) limite com e^{-2x}

$$\lim_{x \rightarrow +\infty} \frac{\frac{2x}{e^{2x} - 1}}{\frac{1}{e^{2x}}} = \lim_{x \rightarrow +\infty} \frac{2x \cdot e^{2x}}{e^{2x} - 1} \stackrel{\text{R. Cauchy}}{=} \lim_{x \rightarrow +\infty} \frac{2 \cdot 2 \cdot e^{2x}}{2e^{2x}} = \lim_{x \rightarrow +\infty} \frac{4e^{2x}}{2e^{2x}} = \lim_{x \rightarrow +\infty} 2 \text{ conv.}$$

como e^{-2n} é convergente (do tipo $e^{\beta n}$ com $\beta < 0$) e o limite $\in \mathbb{R}^+$

$$\int_1^{+\infty} \frac{2n}{e^{2n}-1} dn \text{ também converge}$$

$$c) \int_1^{+\infty} \frac{5n^2-3}{n^8+n-1} dn \rightarrow g(n) = \frac{n^2}{n^8} = \frac{1}{n^6} \text{ (do tipo } \frac{1}{n^\alpha}, \text{ com } \alpha > 0, \text{ logo conv)}$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{5n^2-3}{n^8+n-1}}{\frac{1}{n^6}} = \lim_{n \rightarrow +\infty} \frac{5n^2-3}{n^8+n-1} \cdot n^6 = \lim_{n \rightarrow +\infty} \frac{5n^8-3n^6}{n^8+n-1} = 5$$

$$L = 5 \text{ e } g(n) \text{ conv, logo } \int_1^{+\infty} \frac{5n^2-3}{n^8+n-1} \text{ também converge}$$

$$d) \int_0^{+\infty} e^{n^2} dn \quad e^{n^2} < e^n \text{ (} e^{\beta n} \text{ com } \beta = 0 \text{ logo div)}$$

$$L = \lim_{n \rightarrow +\infty} \frac{e^{n^2}}{e^n} = \lim_{n \rightarrow +\infty} e^{n^2-n} = \lim_{n \rightarrow +\infty} e^{n(n-1)} = +\infty$$

$$\text{como } L = +\infty \text{ e } g(n) \text{ diverge ent } \int_0^{+\infty} e^{n^2} \text{ também diverge}$$

$$e) \int_2^{+\infty} \frac{1}{x^2(1+e^x)} dx$$

$$g(x) = \frac{1}{x^2} \left(\frac{1}{x^\alpha}, \text{ com } \alpha > 1, \text{ logo conv} \right)$$

$$L = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x^2(1+e^x)}}{\frac{1}{x^2}} = \lim_{x \rightarrow +\infty} \frac{x^2}{x^2(1+e^x)} = \lim_{x \rightarrow +\infty} \frac{1}{1+e^x} = 0$$

Pelo C. limite como $L = 0$ e $g(x)$ conv. então $\int_2^{+\infty} \frac{1}{x^2(1+e^x)} dx$ conv.

$$f) \int_1^{+\infty} \frac{x+2}{2x^{5/3}} dx$$

$$g(x) = \frac{1}{x^{5/3}} \left(\frac{1}{x^\alpha}, \alpha > 1, \text{ logo conv.} \right)$$

$$\frac{x+2}{2x^{5/3}} > \frac{1}{2x^{5/3}} \underset{\downarrow \text{conv}}{<} \frac{1}{x^{5/3}} \underset{\text{conv}}{<}$$

$$g(x) = \frac{x}{x^{5/3}} = \frac{1}{x^{2/3}} \text{ (div)}$$

$$L = \lim_{x \rightarrow +\infty} \frac{\frac{x+2}{2x^{5/3}}}{\frac{1}{x^{2/3}}} = \lim_{x \rightarrow +\infty} \frac{x^{2/3}(x+2)}{2x^{5/3}} = \lim_{x \rightarrow +\infty} \frac{x+2}{2x^{-1}} = \lim_{x \rightarrow +\infty} \frac{1}{-2x^{-2}} = \lim_{x \rightarrow +\infty} \frac{x^2}{-2} = +\infty$$

$$\frac{x^{2/3}}{x^{5/3}} = x^{2/3 - 5/3} = x^{-3/3} = x^{-1}$$

$$L = +\infty \text{ e } g(x) \text{ div logo } \int_1^{+\infty} \frac{x+2}{2x^{5/3}} dx \text{ div}$$

$$g) \int_1^{+\infty} \frac{n^2}{(2+n^2)^2} dn = \int_1^{+\infty} \frac{n^2}{4+2n^2+n^4} dn \quad g(n) = \frac{n^2}{n^4} = \frac{1}{n^2} \text{ (conv)}$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{n^2}{4+2n^2+n^4}}{\frac{1}{n^2}} = \lim_{n \rightarrow +\infty} \frac{n^4}{4+2n^2+n^4} \stackrel{\text{Cauchy}}{=} \lim_{n \rightarrow +\infty} \frac{4n^3}{4n+4n^3} \stackrel{\text{Cauchy}}{=} \lim_{n \rightarrow +\infty} \frac{12n^2}{4+12n^2} \stackrel{\text{Cauchy}}{=} \lim_{n \rightarrow +\infty} \frac{24n}{24n} = 1$$

$L \in \mathbb{R}$ e $g(n)$ conv, logo $\int_1^{+\infty} \frac{n^2}{(2+n^2)^2} dn$ converge

$$h) \int_1^{+\infty} \frac{1 - n \sin(n)}{n^3} dn$$

$$\left| \frac{1 - n \sin(n)}{n^3} \right| \leq \frac{1+n}{n^3} < \frac{1}{n^3} \text{ (do tipo } \frac{1}{n^\alpha}, \text{ com } \alpha=3, \text{ logo conv.)}$$

como $\frac{1}{n^3}$ conv, então $\int_1^{+\infty} \frac{1 - n \sin(n)}{n^3} dn$ converge absolutamente (e é convergente)

$$i) \int_2^{+\infty} \frac{3n^2 - 2}{n^3 + 2n + 1} dn \quad g(n) = \frac{n^2}{n^3} = \frac{1}{n} \left(\frac{1}{n^\alpha}, \text{ com } \alpha=1, \text{ logo div} \right)$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{3n^2 - 2}{n^3 + 2n + 1}}{\frac{1}{n}} = \lim_{n \rightarrow +\infty} \frac{n(3n^2 - 2)}{n^3 + 2n + 1} = \lim_{n \rightarrow +\infty} \frac{3n^3 - 2n}{n^3 + 2n + 1} = 3$$

como $L \in \mathbb{R}^+$ e $g(n)$ conv. então o integral também diverge

$$j) \int_1^{+\infty} \frac{\arctg(3n-1)}{1+n^6} dn \quad \frac{-\pi/2}{1+n^6} \leq \frac{\arctg(3n-1)}{1+n^6} \leq \frac{\pi/2}{1+n^6}$$

$$\left| \frac{\arctg(3n-1)}{1+n^6} \right| \leq \frac{\pi}{2+2n^6}$$

$$g(n) = \frac{1}{n^6} \text{ (conv.)}$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{\pi}{2+2n^6}}{\frac{1}{n^6}} = \lim_{n \rightarrow +\infty} \frac{\pi \cdot n^6}{2+2n^6} \stackrel{\text{Cauchy}}{=} \lim_{n \rightarrow +\infty} \frac{6n^5}{12n^5} = \frac{1}{2}$$

como $L \in \mathbb{R}^+$ e $g(n)$ conv. então $\int_1^{+\infty} \frac{\pi}{2+2n^6} dn$ converge então
pelo T. Comparação $\int_1^{+\infty} \frac{\arctg(3n-1)}{1+n^6} dn$ também converge

$$k) \int_1^{+\infty} \frac{e^n}{n} dn < e^n (e^{\beta n} \text{ com } \beta \geq 0 \text{ então div}) \text{ Nada se pode concluir} \\ > \frac{1}{n} \left(\frac{1}{n^\alpha}, \text{ com } \alpha \leq 1 \text{ então div} \right) \text{ então pelo T. Comparação } \int_1^{+\infty} \frac{e^n}{n} dn \text{ também div.}$$

$$1) \int_{-\infty}^{-1} \frac{\sin(n)}{\sqrt[3]{n^5}} dn = - \int_{-1}^{-\infty} \frac{\sin(n)}{\sqrt[3]{n^5}} dn \quad \begin{matrix} \text{M.V. } t = -n \\ dn = -dt \end{matrix} \rightarrow - \int_1^{+\infty} \frac{\sin(-t)}{\sqrt[3]{(-t)^5}} (-dt)$$

$$= \int_1^{+\infty} \frac{\sin(-t)}{(-t)^{5/3}} dt$$

$$\left| \frac{\sin(-t)}{(-t)^{5/3}} \right| \leq \frac{1}{(-t)^{5/3}} < \frac{1}{t^{5/3}}$$

$\uparrow$ conv $\uparrow$ conv $\uparrow$ conv
 $\swarrow$ etão $\swarrow$ etão

pelo T. Comparação $\int_{-\infty}^{-1} \frac{\sin(n)}{\sqrt[3]{n^5}} dn$ é convergente (absolutamente)

7) b) $\int_0^{+\infty} e^{-2n} \underbrace{\sin(\sqrt{n})}_{0 \rightarrow 1} dn < e^{-2n} \text{ (conv)} \text{ pelo C. Comparação conv.}$

$$c) \int_0^{+\infty} \frac{ne^{-n}}{\sqrt{n^2+n+1}} dn > \frac{ne^{-n}}{\sqrt{n}} > \frac{1}{\sqrt{n}} \text{ (div)}$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{ne^{-n}}{\sqrt{n^2+n+1}}}{\frac{1}{\sqrt{n}}} = \lim_{n \rightarrow +\infty} \frac{\sqrt{n} \cdot n \cdot e^{-n}}{\sqrt{n^2+n+1}} \stackrel{\text{Cauchy}}{=} \lim_{n \rightarrow +\infty} \frac{\frac{1}{2} n^{-1/2} \cdot 1 \cdot -e^{-n}}{\frac{1}{2} (2n+1)^{-1/2}} = \lim_{n \rightarrow +\infty} \frac{\frac{1}{2\sqrt{n}} \cdot -e^{-n}}{\frac{1}{2\sqrt{2n+1}}} = \lim_{n \rightarrow +\infty} \frac{-e^{-n} \cdot 2\sqrt{2n+1}}{2\sqrt{n}}$$

$$= +\infty \quad \text{div}$$

6) a) $\int_1^{+\infty} \frac{\sin^2 n}{n^{5/2}} < \frac{1}{n^{5/2}}$ e' do tip. $\frac{1}{x^\alpha}$ com $\alpha > 1$ (conv.), logo $\int_1^{+\infty}$ e' conv.

b) $\int_1^{+\infty} \frac{2x}{e^{2x}-1} dx$

$$\frac{2x}{e^{2x}-1} < \frac{1}{e^{2x}-1}$$

$$\frac{\frac{1}{e^{2x}-1}}{\frac{1}{e^{2x}}} = \frac{e^{2x}}{e^{2x}-1} = \frac{1}{1-\frac{1}{e^{2x}}} = 1$$

$$c) \int_1^{+\infty} \frac{5x^2 - 3}{x^8 + x - 1} dx$$

$$g(x) = \frac{x^2}{x^8} = \frac{1}{x^6} \quad (\text{conv.})$$

$$L = \lim_{x \rightarrow +\infty} \frac{\frac{5x^2 - 3}{x^8 + x - 1}}{\frac{1}{x^6}} =$$

$$= \lim_{x \rightarrow +\infty} \frac{x^6(5x^2 - 3)}{x^8 + x - 1} = \lim_{x \rightarrow +\infty} \frac{5x^8 - 3x^6}{x^8 + x - 1} = \frac{5 - \frac{3}{x^2}}{1 + \frac{1}{x^7} - \frac{1}{x^8}} = 5$$

como $L \in \mathbb{R}^+$ e $g(x)$ é convergente então o integral é conv.

$$d) \int_0^{+\infty} e^{x^2} dx$$

$$e^{x^2} > e^x$$

$$h) \int_1^{+\infty} \frac{1 - x \sin x}{x^3} dx$$

$$-1 \leq \sin x \leq 1$$

$$0 \leq \sin x \leq 1$$

$$g(x) = \frac{1}{x^3} \quad (\text{conv.})$$

$$\left| \frac{1 - x \sin x}{x^3} \right| \leq \frac{1 + x}{x^3} < \frac{1}{x^3}$$

$$a) \int_2^{+\infty} \frac{1}{\sqrt{n^5+2n}} \, dn = \frac{1}{(n^5+2n)^{1/2}} \quad g(n) = \frac{1}{n^{1/2}} \quad \left(\frac{1}{n^\alpha}, \text{con } \alpha < 1, \text{div} \right)$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{1}{(n^5+2n)^{1/2}}}{\frac{1}{n^{1/2}}} = \lim_{n \rightarrow +\infty} \frac{n^{1/2}}{(n^5+2n)^{1/2}} = \lim_{n \rightarrow +\infty} \sqrt{\frac{n}{n^5+2n}} = \lim_{n \rightarrow +\infty} \sqrt{\frac{1}{n^4+2}} = 0 \quad \text{Nada se pode concluir}$$

$$\frac{1}{\sqrt{n^5+2n}} > \frac{1}{n^5+2n} > \frac{1}{n^5}$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{1}{\sqrt{n^5+2n}}}{\frac{1}{n^5}} = \lim_{n \rightarrow +\infty} \frac{n^5}{\sqrt{n^5+2n}} \stackrel{\text{Cauchy}}{=} \lim_{n \rightarrow +\infty} \frac{5n^4}{\frac{1}{2}(n^5+2n)^{-1/2} \cdot 5n^4+2} = \lim_{n \rightarrow +\infty} \frac{5n^4 \sqrt{n^5+2n}}{\frac{5n^4}{2}+2}$$

$$= \lim_{n \rightarrow +\infty} \frac{2(5n^4 \sqrt{n^5+2n})}{5n^4+2} = 2 \lim_{n \rightarrow +\infty} \frac{5n^4 \cdot n^5+2n}{5n^4+2} = \infty. \quad \text{Nada}$$

$$\int_2^{+\infty} \frac{1}{\sqrt{n^5+2n}} \, dn < \frac{1}{\sqrt{n^5}} = \frac{1}{n^{5/2}} \quad \leftarrow \text{conv}$$

$$b) \int_0^{+\infty} e^{-2n} \sin \sqrt{n} \, dn \quad \left| e^{-2n} \sin \sqrt{n} \right| \leq e^{-2n} \quad \left(e^{\beta n} \text{ con } \beta < 0, \text{conv} \right)$$

Pelo c. Comparação $\int_0^{+\infty} e^{-2n} \sin \sqrt{n} \, dn$ conv.

$$\textcircled{9} \quad f(s) = \mathcal{L}\{f(t)\}(s) = ?$$

$$a) \quad f(t) = 2 \sin(3t) + t - 5e^{-t}$$

$$2\mathcal{L}\{\sin(3t)\} + \mathcal{L}\{t\} - 5\mathcal{L}\{e^{-t}\} = 2 \frac{3}{s^2 + 9} + \frac{1}{s^2} - 5 \frac{1}{s+1}, \quad s > 0$$

$$b) \quad f(t) = e^{-t} \sinh(5t) = \mathcal{L}\{\sinh(5t)\}(s+1) = \frac{5}{(s+1)^2 - 25}, \quad s > 6$$

$$c) \quad f(t) = e^{2t} \cos(5t) = \mathcal{L}\{\cos(5t)\}(s-2) = \frac{s-2}{(s-2)^2 + 25}, \quad s > 2$$

$$d) \quad f(t) = t e^{3t} = \mathcal{L}\{t\}(s-3) = \frac{1}{(s-3)^2}, \quad s > 3$$

$$= (-1)^1 \left(\mathcal{L}\{e^{3t}\} \right)'(s) = -1 \left(\frac{1}{s-3} \right)' = \frac{1}{(s-3)^2}, \quad s > 3$$

$$e) \quad f(t) = t \cosh(t) = (-1)^1 \left(\mathcal{L}\{\cosh(t)\} \right)'(s) = -1 \left(\frac{s}{s^2 - 1} \right)' \quad s > 1$$

$$= - \left(\frac{1(s^2 - 1) - (s^2 - 1)' \cdot s}{(s^2 - 1)^2} \right) = - \frac{s^2 - 1 - 2s \cdot s}{(s^2 - 1)^2}$$

$$= - \frac{s^2 - 1 - 2s^2}{(s^2 - 1)^2} = - \frac{-s^2 - 1}{(s^2 - 1)^2}, s > 1$$

$$f) f(t) = \pi - 5e^{-t} \cdot t^{10} = \mathcal{L}\{\pi\} - 5 \mathcal{L}\{t^{10}\}(s+1) = \frac{\pi}{s} - 5 \frac{10!}{(s+1)^{11}}, s > 0$$

$$g) f(t) = (3t - 1) \sin t = 3t \sin t - \sin t = 3 \mathcal{L}\{t \sin t\} - \mathcal{L}\{\sin t\} =$$

$$= 3 \left((-1)^1 \left(\frac{1}{s^2 + 1} \right)' \right) - \frac{1}{s^2 + 1} = 3 \frac{2s}{(s^2 + 1)^2} - \frac{1}{s^2 + 1} = \frac{6s}{(s^2 + 1)^2} - \frac{1}{s^2 + 1}$$

$$h) f(t) = (1 - H_{\pi}(t)) \sin t = \sin t - H_{\pi}(t) (\sin t)$$

$$\mathcal{L}\{\sin t\} - \mathcal{L}\{H_{\pi}(t) \cdot \sin t\}(s) = \frac{1}{s^2 + 1} - \mathcal{L}\{-H_{\pi}(t) \cdot \sin(t - \pi)\}(s)$$

$$= \frac{1}{s^2 + 1} + e^{-\pi s} \mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1} + e^{-\pi s} \frac{1}{s^2 + 1} = \frac{1}{s^2 + 1} + \frac{e^{-\pi s}}{s^2 + 1}, s > 0$$

$$i) f(t) = (t-2)^2 e^{2(t-2)} H_2(t) = H_2(t) \cdot (t-2)^2 \cdot e^{2t-4} =$$

$$f(t) = t^2 e^{2t}$$

$$(t-2)^2 \cdot e^2 \cdot e^{(t-2)} = (e(t-2))^2 \cdot e^2 = (et - 2e)^2 \cdot e^2 =$$

??

$$11) \quad a) \int_0^{+\infty} t^{10} e^{-2t} dt = \mathcal{L}\{t^{10}\}(s=2) = \frac{10!}{2^{11}}$$

$$\mathcal{L}\{f(t)\}(s) = \int_0^{+\infty} e^{-st} f(t) dt$$

$$b) \int_0^{+\infty} e^{-3t} t \cdot \sin t dt = \mathcal{L}\{t \sin(t)\}(3) = (-1) \left(\mathcal{L}\{\sin t\} \right)'(3) = - \left(\frac{1}{s^2 + 1} \right)' \Big|_{s=3} \\ = - \frac{-(s^2 + 1)'}{(s^2 + 1)^2} \Big|_{s=3} = \frac{2s}{(s^2 + 1)^2} \Big|_{s=3} = \frac{6}{10^2} = \frac{6}{100} = \frac{3}{50}$$

$$(10) \quad \mathcal{L}^{-1}\{f(s)\}(t)$$

$$a) f(s) = \frac{2s}{s^2 - 9}$$

$$\mathcal{L}^{-1}\{f(s-\lambda)\} = e^{\lambda t} \mathcal{L}^{-1}\{f(s)\}$$

$$\mathcal{L}^{-1}\{f(s)\}(t) = f(t)$$

$$\mathcal{L}^{-1}\left\{\frac{2s}{s^2 - 3^2}\right\}(t) = 2 \mathcal{L}^{-1}\left\{\frac{s}{s^2 - 3^2}\right\}(t) = 2 \cosh(3t)$$

$$b) f(s) = \frac{4}{s^7}$$

$$\mathcal{L}^{-1} \left\{ \frac{4}{s^7} \right\} = 4 \mathcal{L}^{-1} \left\{ \frac{1}{s^7} \right\} = \frac{4}{6!} \mathcal{L}^{-1} \left\{ \frac{6!}{s^7} \right\} = \frac{4}{6!} t^6 = \frac{4}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2} t^6 = \frac{t^6}{30 \cdot 6} = \frac{t^6}{180}$$

$$c) f(s) = \frac{1}{s^2 + 6s + 9}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 6s + 9} \right\} (t) = \mathcal{L}^{-1} \left\{ \frac{1}{(s+3)^2} \right\} = e^{-3t} \cdot \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} = e^{-3t} \cdot t$$

$$d) f(s) = \frac{1}{s^2 + s - 2}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + s - 2} \right\}$$

$$C.1 \quad \frac{1}{s^2 + s - 2} = \frac{1}{(s+2)(s-1)} = \frac{A}{s+2} + \frac{B}{s-1} = \frac{A(s-1) + B(s+2)}{(s+2)(s-1)}$$

$$\begin{cases} A + B = 0 \\ -A + 2B = 1 \end{cases} \quad \begin{cases} A = -B \\ 3B = 1 \end{cases} \quad \begin{cases} A = -1/3 \\ B = 1/3 \end{cases}$$

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + s - 2} \right\} (t) &= \mathcal{L}^{-1} \left\{ \frac{-1/3}{s+2} \right\} + \mathcal{L}^{-1} \left\{ \frac{1/3}{s-1} \right\} = \\ &= -\frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\} + \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{1}{s-1} \right\} = -\frac{1}{3} e^{-2t} + \frac{1}{3} e^{1t} = \frac{e^t - e^{-2t}}{3} \end{aligned}$$

$$e) f(s) = \frac{1}{s^2 + 4s + 6} = \frac{1}{s^2 + 4s + 4 + 2} = \frac{1}{(s+2)^2 + 2} = \frac{1}{(s+2)^2 + \sqrt{2}^2}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s+2)^2 + \sqrt{2}^2} \right\} = e^{-2t} \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + \sqrt{2}^2} \right\} = e^{-2t} \cdot \frac{1}{\sqrt{2}} \mathcal{L}^{-1} \left\{ \frac{\sqrt{2}}{s^2 + \sqrt{2}^2} \right\} = \frac{e^{-2t}}{\sqrt{2}} \sin(\sqrt{2}t), t \geq 0$$

$$f) f(s) = \frac{3s - 1}{s^2 - 4s + 13} = \frac{3s - 1}{s^2 - 4s + 4 + 9} = \frac{3s - 1}{(s-2)^2 + 9} = \frac{3s - 1}{(s-2)^2 + 3^2}$$

$$\mathcal{L}^{-1} \left\{ \frac{3s - 1}{(s-2)^2 + 9} \right\} = \mathcal{L}^{-1} \left\{ \frac{3s}{(s-2)^2 + 9} \right\} - \mathcal{L}^{-1} \left\{ \frac{1}{(s-2)^2 + 9} \right\} = " - e^{2t} \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 3^2} \right\}$$

$$= " - \frac{e^{2t}}{3} \mathcal{L}^{-1} \left\{ \frac{3}{s^2 + 3^2} \right\} = " - \frac{e^{2t}}{3} \sin(3t) = *$$

C.A

$$\mathcal{L}^{-1} \left\{ \frac{3s}{(s-2)^2 + 9} \right\} = e^{2t} \mathcal{L}^{-1} \left\{ \frac{3(s+2)}{s^2 + 3^2} \right\} = e^{2t} \mathcal{L}^{-1} \left\{ \frac{3s + 6}{s^2 + 3^2} \right\} =$$

$$= e^{2t} \mathcal{L}^{-1} \left\{ \frac{3s}{s^2 + 3^2} \right\} + e^{2t} \mathcal{L}^{-1} \left\{ \frac{6}{s^2 + 3^2} \right\} = 3e^{2t} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 3^2} \right\} + 2e^{2t} \mathcal{L}^{-1} \left\{ \frac{3}{s^2 + 3^2} \right\}$$

$$= 3e^{2t} \cos(3t) + 2e^{2t} \sin(3t)$$

$$* = 3e^{2t} \cos(3t) + 2e^{2t} \sin(3t) - \frac{e^{2t}}{3} \sin(3t) = e^{2t} \left(3\cos(3t) + 2\sin(3t) - \frac{\sin(3t)}{3} \right)$$

$$= e^{2t} \left(\frac{9 \cos(3t) + 6 \sin(3t) - \sin(3t)}{3} \right) = e^{2t} \left(\frac{9 \cos(3t) + 5 \sin(3t)}{3} \right) =$$

$$= e^{2t} \left(3 \cos(3t) + \frac{5}{3} \sin(3t) \right), \quad t \geq 0$$

11)

a) $\int_0^{+\infty} t^{10} e^{-2t} dt = ?$

$$\mathcal{L}\{f(t)\}(s) = \int_0^{+\infty} e^{-st} f(t) dt \Rightarrow \mathcal{L}\{t^{10}\}(2) = \frac{10!}{2^{11}}$$

b) $\int_0^{+\infty} e^{-3t} \cdot t \cdot \sin t \, dt$

$$\mathcal{L}\{f(t)\}(s) = \int_0^{+\infty} e^{-st} f(t) dt \rightarrow \mathcal{L}\{t \sin(t)\}(3) = (-1)^1 \mathcal{L}\{\sin(t)\}'(3) = (-1) \left(\frac{1}{s^2 + 1} \right)' \Big|_{s=3}$$

$$= - \frac{-2s}{(s^2 + 1)^2} \Big|_{s=3} = \frac{6}{(10)^2} = \frac{6}{100} = \frac{3}{50}$$

12) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f'(t) + 2f(t) = e^t$, $f(0) = 2$. $f(t) = ?$

$$\mathcal{L}\{f'(t) + 2f(t)\} = \mathcal{L}\{e^t\} \Leftrightarrow \mathcal{L}\{f'(t)\} + 2\mathcal{L}\{f(t)\} = \frac{1}{s-1} \quad (=)$$

$$\Leftrightarrow s^1 \mathcal{L}\{f(t)\} - s^0 f(0) + 2\mathcal{L}\{f(t)\} = \frac{1}{s-1} \quad (=)$$

$$\Rightarrow sL\{f(t)\} + 2L\{f(t)\} = \frac{1}{s-1} + f(0) \Rightarrow$$

$$\Leftrightarrow (s+2)L\{f(t)\} = \frac{1}{s-1} + 2 \Rightarrow (s+2)L\{f(t)\} = \frac{1+2s-2}{s-1} \Rightarrow$$

$$\Rightarrow L\{f(t)\} = \frac{2s-1}{(s+2)(s-1)} \Rightarrow f(t) = L^{-1}\left\{\frac{2s-1}{(s+2)(s-1)}\right\}$$

$$L^{-1}\left\{\frac{2s-1}{(s+2)(s-1)}\right\} = 2L^{-1}\left\{\frac{s}{(s+2)(s-1)}\right\} - L^{-1}\left\{\frac{1}{(s+2)(s-1)}\right\} = *$$

CA

$$\frac{s}{(s+2)(s-1)} = \frac{A}{s+2} + \frac{B}{s-1} = \frac{A(s-1) + B(s+2)}{(s+2)(s-1)}$$

$$\begin{cases} A+B=1 \\ -A+2B=0 \end{cases} \begin{cases} A=-B+1 \\ -B+2B=-1 \end{cases} \begin{cases} A=2 \\ B=-1 \end{cases}$$

CA

$$\frac{1}{(s+2)(s-1)} = \frac{A}{s+2} + \frac{B}{s-1} = \frac{A(s-1) + B(s+2)}{(s+2)(s-1)}$$

$$\begin{cases} A+B=0 \\ -A+2B=1 \end{cases} \begin{cases} A=-B \\ 3B=1 \end{cases} \begin{cases} A=-1/3 \\ B=1/3 \end{cases}$$

$$* = 2L^{-1}\left\{\frac{2}{s+2} + \frac{-1}{s-1}\right\} - L^{-1}\left\{\frac{-1/3}{s+2} + \frac{1/3}{s-1}\right\}$$

$$= 2L^{-1}\left\{\frac{2}{s+2}\right\} - 2L^{-1}\left\{\frac{1}{s-1}\right\} + \frac{1}{3}L^{-1}\left\{\frac{1}{s+2}\right\} + \frac{1}{3}L^{-1}\left\{\frac{1}{s-1}\right\}$$

$$= 4L^{-1}\left\{\frac{1}{s+2}\right\} - 2e^{1t} + \frac{1}{3}e^{-2t} + \frac{1}{3}e^t$$

$$= 4e^{-2t} - 2e^t + \frac{1}{3}e^{-2t} + \frac{1}{3}e^t = \frac{12e^{-2t} - 6e^t + e^{-2t} + e^t}{3} = \frac{13e^{-2t} - 5e^t}{3} =$$

Exs revisão

(13) Natureza de:

$$\int_1^{+\infty} \frac{1}{(1+u^2) \operatorname{arctg} u} du = \lim_{t \rightarrow +\infty} \int_1^t \frac{1}{1+u^2} \cdot \frac{1}{\operatorname{arctg} u} du$$

$u = \operatorname{arctg} u$
 $du = \frac{1}{1+u^2} du$

$$= \lim_{t \rightarrow +\infty} \int_1^t \frac{1}{u} du = \lim_{t \rightarrow +\infty} [\ln|u|]_1^t =$$

$$= \lim_{t \rightarrow +\infty} [\ln|\operatorname{arctg} u|]_1^t = \lim_{t \rightarrow +\infty} (\ln|\operatorname{arctg}(t)| + \ln|\operatorname{arctg}(1)|)$$

$$= \ln\left(\frac{\pi}{2}\right) + \ln\left(\frac{\pi}{4}\right) = \ln\left(\frac{\frac{\pi}{2}}{\frac{\pi}{4}}\right) = \ln\left(\frac{4\pi}{2\pi}\right) = \ln(2) + c, \quad c \in \mathbb{Z}$$

integral convergente

(14) Natureza de:

$$\int_{-\infty}^0 e^u (4-u) du = - \int_0^{-\infty} e^u (4-u) du = - \int_0^{+\infty} e^{-t} (4+t) (-1) dt = \int_0^{+\infty} e^{-t} (4+t) dt$$

$$e^{-t} (4+t) < e^{-t} \quad (\text{do tipo } e^{\beta t} \text{ com } \beta < 0 \rightarrow \text{conv})$$

Pelo C. Comparação $\int_{-\infty}^0 e^n (4-n) \, dn$ é convergente

$$\int_{-\infty}^0 e^n (4-n) \, dn = \lim_{t \rightarrow -\infty} \int_t^0 e^n (4-n) \, dn = \lim_{t \rightarrow -\infty} \left((4-n) e^n \Big|_t^0 - \int_t^0 e^n \cdot (-1) \, dn \right)$$

$$\begin{aligned} u &= (4-n) & dv &= e^n \\ du &= -1 \, dn & v &= e^n \end{aligned}$$

$$= \lim_{t \rightarrow -\infty} \left[(4-n) e^n + e^n \right]_t^0 = \lim_{t \rightarrow -\infty} (4 \cdot 1 + 1 - ((4-t) \cdot 0 + 0)) = 5 - 0 = 5$$

(15) $f: [1, +\infty] \rightarrow \mathbb{R}$, $f(n) = \frac{1}{2\sqrt{\ln n}}$

a) primitiva de f que se anula em $n=e$?

$$\int \frac{1}{2\sqrt{\ln n}} \, dn = \int \frac{(\ln n)^{-1/2}}{n} \, dn = \frac{(\ln n)^{1/2}}{\frac{1}{2}} + C = 2\sqrt{\ln(n)} + C$$

$$2\sqrt{\ln(e)} + C = 0 \Leftrightarrow 2\sqrt{1} = -C \Leftrightarrow C = -2$$

$$\therefore 2\sqrt{\ln(n)} - 2$$

b) Natureza de $\int_2^{+\infty} \frac{1}{n\sqrt{\ln n}} \, dn$

$$g(n) = \frac{1}{n} \text{ (do tipo } \frac{1}{n^\alpha} \text{ com } \alpha = 1, \text{ div)}$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{1}{n \sqrt{\ln n}}}{\frac{1}{n}} = \lim_{n \rightarrow +\infty} \frac{n}{n \sqrt{\ln n}} = \lim_{n \rightarrow +\infty} \frac{1}{\sqrt{\ln n}} = \frac{1}{\infty} = 0 \quad \text{Nada se pode concluir}$$

$$\frac{1}{n \sqrt{\ln n}} > \frac{1}{\sqrt{\ln n}} > \frac{1}{\sqrt{n}}$$

$\uparrow \text{div} \quad \quad \uparrow \text{div} \quad \quad \uparrow \text{div}$

Pelo C. Comparação o integral é divergente

(18)

$$a) \mathcal{L} \{ e^{-3t} \sin(2t) \} = \mathcal{L} \{ \sin(2t) \} (s+3) = \frac{2}{(s+3)^2 + 2^2}, \quad s > -3 = \frac{2}{s^2 + 6s + 13}, \quad s > -3$$

$$b) \mathcal{L} \{ (t - 2 + e^{-2t}) \cos(4t) \} = \mathcal{L} \{ t \cos(4t) \} - 2 \mathcal{L} \{ \cos(4t) \} + \mathcal{L} \{ \cos(4t) \} (s+2)$$

$$= -1 \left(\frac{s}{s^2 + 4^2} \right)'(s) - 2 \left(\frac{s}{s^2 + 4^2} \right) + \frac{s+2}{(s+2)^2 + 4^2} =$$

$$= - \left(\frac{s^2 + 16 + 2s \cdot s}{(s^2 + 4^2)^2} \right) - \frac{2s}{s^2 + 16} + \frac{s+2}{s^2 + 4s + 20} = - \frac{3s^2 + 16}{(s^2 + 4^2)^2} - \frac{2s}{s^2 + 16} + \frac{s+2}{s^2 + 4s + 20}, \quad s > 0$$

(19)

$$\int_0^{+\infty} t \cos(t) e^{2t} dt = ?$$

$$\int_0^{+\infty} e^{-st} f(t) dt = \mathcal{L} \{ f(t) \}(s)$$

$$\int_0^{+\infty} e^{2t} \cdot t \cos(t) dt = \mathcal{L}\{t \cdot \cos(t)\}(2) = -1 \cdot \left(\frac{s}{s^2+1^2}\right)' \Big|_{s=2} = -\frac{s^2+1-2s \cdot s}{(s^2+1)^2} \Big|_{s=2}$$

$$= -\frac{2^2+1-2 \cdot 2 \cdot 2}{5^2} = -\frac{5-8}{25} = -\frac{-3}{25} = \frac{3}{25}$$

2) c) $\int_{-\infty}^{+\infty} 2^n du = \int_{-\infty}^0 2^n du + \int_0^{+\infty} 2^n du$

CA $\int_{-\infty}^0 2^n du = \lim_{t \rightarrow -\infty} \int_t^0 2^n du = \lim_{t \rightarrow -\infty} \left[\frac{2^n}{\ln|2|} \right]_t^0 = \lim_{t \rightarrow -\infty} \frac{2^0}{\ln(2)} - \frac{2^t}{\ln 2} = \frac{1}{\ln 2} - \frac{0}{\ln(2)} = \frac{1}{\ln 2}$

$$\int_0^{+\infty} 2^n du = \lim_{t \rightarrow +\infty} \left[\frac{2^n}{\ln 2} \right]_0^t = \lim_{t \rightarrow +\infty} \frac{2^t}{\ln 2} - \frac{1}{\ln 2} = +\infty \text{ div} \rightarrow \text{então } \int_{-\infty}^{+\infty} 2^n du \text{ é div}$$

Natureza de

a) $\int_1^{+\infty} \frac{\sin^2(n)}{n^{5/2}}$

$$\frac{\sin^2 n}{n^{5/2}} \leq \frac{1}{n^{5/2}} \quad (\text{conv pq } \alpha > 1) \quad \text{Pelo C. Comparação}$$

$$\int_1^{+\infty} \frac{\sin^2(n)}{n^{5/2}} \text{ conv}$$